Sandeep Malik

AI Architect | Principal Engineer — Agentic AI, LLM Platforms & Harness Engineering
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SUMMARY

AI Architect and hands-on Principal Engineer with 14+ years across distributed systems, data platforms, and AI. I take AI agents from demo to production: the runtime, orchestration, deterministic controls, evals, and observability that let a whole engineering organization rely on them. At Autodesk I lead a team of 4 building the internal platform teams use to create and run agents and MCP servers (3 production agents, 5 MCP servers, 2 IDE plugins), partner with our Distinguished Architect on its architecture, and set how the team builds with AI: a shared agentic SDLC, evals plus Dynatrace APM monitoring, and coaching on tactical vs. strategic programming. Flagship: an agent that migrated 1,500+ internal applications to GraphQL with over 85% success. Built on a decade of data platforms: ~200K events/sec search, ~2 PB pipelines, and Databook's first semantic layer.

CORE SKILLS

Agentic AI & Architecture: AI agents, LLM platforms, enterprise AI solutions, agent harnesses and runtimes, multi-agent systems and orchestration, MCP interoperability, tool integration, context assembly, RAG, deterministic workflows, human oversight
AI Quality & Trust: Agentic SDLC standards, AI-assisted code quality, agent evaluation, LLM-as-judge, guardrails, AI observability (evals, Dynatrace APM, Opik), secure coding, RBAC, vulnerability and compliance data, production monitoring
Models & Frameworks: LangGraph, Google ADK, LangChain, LlamaIndex, Azure OpenAI, AWS Bedrock, Pinecone, Elasticsearch, OpenSearch
Cloud, Data & Engineering: Python, Java, Scala, SQL, Node.js, Snowflake, Databricks, Spark, Kafka, PostgreSQL, AWS, Azure, Kubernetes, Docker, Terraform, CI/CD

EXPERIENCE

Principal Software Engineer / Technical Lead, Autodesk, Trust AI Engineering

Jul 2025 – Present | Montreal, Canada

- Lead a team of 4 building Autodesk's central platform for AI agents and MCP servers (shared catalog, discovery, orchestration, and runtime) so any team can build and run agents safely; work closely with the Distinguished Architect on platform design.
- Migrated 1,500+ internal applications off legacy, vulnerability-prone APIs to GraphQL with an agent I built (Python, Go, Java, Ruby on Rails, Perl) at greater than 85% success by code complexity; the AI works inside a controlled environment and deterministic checks decide what ships, with progress tracked in a UI dashboard.
- Shipped 3 production agents, 5 MCP servers, and 2 IDE plugins used across Autodesk; built a reusable agent runtime (shared auth, typed outputs, guards, observability) so teams assemble agents from common parts.
- Created one shared AI development workflow (intent, design, plan, ADR, spec, implementation loop, docs) packaged as a plugin of skills and agents built on Autodesk guidelines, so every engineer produces the same artifacts.
- Defined how the team monitors agents and MCP servers: evals for output quality and Dynatrace (APM) for runtime health in production; coach engineers on tactical vs. strategic programming with AI and review the quality of AI-generated code.
- Built security into developers' daily tools: an MCP server that checks code against secure-coding and cloud standards in the IDE, a multi-step threat-modeling workflow, and agents that detect repo vulnerabilities and open fix PRs.

Data Architect / Staff Engineer, Databook Labs

Dec 2022 – Jul 2025 | Montreal, Canada (Remote)

- Built the company data platform from the ground up and delivered its first semantic layer (early 2023), powering Databook AI.
- Led a GPT product for strategic sales using multi-query and step-back retrieval, reranking, and LLM-as-judge to improve grounding and personalization.

- Designed agentic workflows with LangGraph and Google ADK and a semantic index across Pinecone, Elasticsearch, Snowflake, and Databricks; led a team of data engineers.

Principal Data Engineer, Sitecore
Mar 2022 – Dec 2022 | Montreal, Canada
- Led the core Search team supporting AI-backed search and e-commerce personalization at ~200,000 events per second.
- Architected a data lake and warehouse for a global retailer, reducing storage costs; delivered Snowflake marts and automated monitoring.

Solutions Architect, Cloudera
May 2021 – Mar 2022 | Montreal, Canada
- Advised customers adopting Cloudera Data Platform across public, private, and hybrid environments; designed on-premises to AWS migrations and trained engineering teams.

Senior Data Engineer, Intact Insurance
Apr 2020 – May 2021 | Montreal, Canada
- Led data engineers delivering an on-premises to AWS historical-data migration and refactoring pipelines onto Databricks; authored cloud and data-platform best-practice guides.

Earlier: Data Engineer, Ericsson (2019–20) · Data Engineer, CGI (2018–19) · Data Engineer, Synechron (2018) · Big Data Developer, Interglobe (2016–18, Dubai — ~2 PB pipelines) · Software Engineer, MothersonSumi/MIND (2014–16) · Software Developer, NIIT (2012–14).

SELECTED PROJECT

Agentic Expense Manager (open source): Local-first Claude tool-calling agent for web and WhatsApp on a shared service layer with a SQLite source of truth (FastAPI, React); server-side tax calculation, receipt OCR, optional Google Drive/Sheets sync; 345+ backend tests including adversarial sync cases.

CERTIFICATIONS

Snowflake SnowPro Core · AWS Solutions Architect – Associate · AWS Cloud Practitioner · Databricks Certified Associate Developer (Spark 3.0) · Cloudera CCA Spark and Hadoop Developer (CCA-175) · Microsoft Azure Fundamentals (AZ-900)

EDUCATION

Master of Science (M.Sc.), Computer Science — Manav Bharti University (2010–2013)